NexusOS — 3-Axis Independent Bifilar Toroid Driver

Circuit Board Specification · Version 1.0 · July 2026 · wnsp.io

1. Project Overview

Three independently driven bifilar-wound toroid coils (~62.2 ;Ä, V 6,’ &W V— ing three separate driver circuits. Each axis is electrically isolated from the others. A single Si5351A clock generator provides three phase-configurable clock signals — one per axis.

Deliverable: One controller board + three identical driver boards, fully assembled and bench-tested.

2. System Block Diagram

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CONTROLLER BOARD
  Arduino Nano % %  I²C % % % %  Si5351A
                      CLK0 % % !’ Driver Board X  (0°)
                      CLK1 % % !’ Driver Board Y  (120°)
                      CLK2 % % !’ Driver Board Z  (240°)
  12V IN % % % % % % % % % % % % % % % % % % % % % % !’ Shared power rail

DRIVER BOARD (× 3 identical)
  CLK IN % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % !’ IR2104 #1 IN
  CLK IN % % !’ [74HC04 inverter] % % % % % % % !’ IR2104 #2 IN
  IR2104 #1 HO/LO !’ Q1/Q2 !’ Node A !’ Conductor A (+)
  IR2104 #2 HO/LO !’ Q3/Q4 !’ Node B !’ Conductor B (" )
  Tuning Cap C1 (1 nF) across Node A – Node B
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3. Controller Board Specification

Parameter	Value
MCU	Arduino Nano (ATmega328P)
Clock generator	Adafruit Si5351A breakout (25 MHz TCXO)
Interface	I²C — SDA pin A4, SCL pin A5
Supply voltage	5 V USB or regulated 5 V DC
CLK output level	3.3 V logic square wave
CLK frequency	638 kHz (tunable 8 kHz – 150 MHz via firmware)
CLK0 phase	0°
CLK1 phase	120°
CLK2 phase	240°
Output connector	3× JST 2-pin (one per CLK output)
Firmware	Adafruit Si5351 Arduino library — pre-loaded by client

4. Driver Board Specification (Per Axis)

Topology: Full H-bridge (differential drive). Both bifilar conductors driven in anti-phase simultaneously.

• Node A (Conductor A): HIGH when CLK = HIGH

• Node B (Conductor B): HIGH when CLK = LOW

• Net effect: current alternates direction through both conductors at CLK frequency, creating the :„¶³)"—:„!&³) standing wave condition.

Operating Parameters:

Parameter	Value
Drive frequency	638 kHz (set by controller)

Supply voltage

12 V DC

MOSFET gate threshold

< 2 V (logic-level gate required)

Peak coil current

~500 mA (limited by DCR ~0.3 :'•

Tuning capacitor

1 nF ±5% (LC resonance with ~62.2 ;Ä‚•

LC resonant frequency

~638 kHz

Output waveform

Differential square wave, 180° anti-phase

5. Bill of Materials — Per Driver Board (× 3)

Ref	Part	Value / Part#	Qty	Notes
U1	Half-bridge gate driver	IR2104 (DIP-8)	1	
U2	Half-bridge gate driver	IR2104 (DIP-8)	1	
U3	Hex inverter	74HC04 (DIP-14)	1	Anti-phase
Q1,Q3	N-ch MOSFET (high-side)	IRLZ44N preferred	2	Logic-level
Q2,Q4	N-ch MOSFET (low-side)	IRLZ44N preferred	2	Logic-level
C1	Tuning capacitor	1 nF ceramic 50 V	1	LC tank
C2,C3	Bootstrap capacitor	100 nF ceramic 25 V	2	IR2104 HO
C4	Decoupling	10 ;Äb VÆV7G&öÇ—F–2 #R V	1	
R1–R4	Gate resistors	10 :'Â ã#R W	4	Damp ringing
R5	Pull-down	10 k:•	1	CLK line
J1	CLK input connector	JST 2-pin	1	From Si5351
J2	Coil connector	4-pin screw terminal	1	Bifilar A+B
J3	Power connector	2-pin screw terminal	1	12 V + GND
—	PCB	50×70 mm, 2-layer FR4	1	

5b. Controller Board Additional Parts

Part	Qty	Notes
Arduino Nano (ATmega328P)	1	Pre-programmed
Adafruit Si5351A breakout	1	With 25 MHz TCXO
10 k:' VÆÂ×W &W6—7F÷'2 „›$2 Æ–æW2•	2	SDA + SCL
100 nF decoupling capacitor	2	VCC bypass
PCB or half-size breadboard	1	

6. Interface Definitions

CLK Input J1 (per driver board):
  Pin 1 — CLK signal (3.3 V logic, 638 kHz square wave)
  Pin 2 — GND

Coil Connector J2 (per driver board):
  Pin 1 — Conductor A, terminal 1
  Pin 2 — Conductor A, terminal 2
  Pin 3 — Conductor B, terminal 1
  Pin 4 — Conductor B, terminal 2

Power J3 (per driver board):
  Pin 1 — 12 V DC
  Pin 2 — GND

7. Critical Build Notes for Freelancer

1. All three driver boards are IDENTICAL circuits — only the CLK input source differs (CLK0 / CLK1 / CLK2).
2. The 74HC04 inverter provides the anti-phase signal for IR2104 #2 — do not omit.
3. MOSFETs MUST be logic-level gate threshold (Vgs(th) < 2 V). Standard IRF540N is marginal at 3.3 V gate — IRLZ44N strongly preferred.
4. Keep gate drive traces short — 638 kHz switching, minimise stray inductance on gate loops.
5. Bootstrap capacitors (C2, C3) are required for high-side MOSFET turn-on — do not omit.
6. Toroid physical dimensions and lead identification supplied by client on coil delivery.
7. Label completed boards clearly: X-AXIS (CLK0), Y-AXIS (CLK1), Z-AXIS (CLK2).

8. Acceptance / Test Criteria

#	Test	Pass Condition
1	No-load oscillation	Oscilloscope confirms Node A and Node B at 638 kHz, 180° anti-phase
2	Resistive load (10 :’•	Current flows without MOSFET exceeding 40 °C after 5 min continuous
3	Coil load (actual toroid)	Oscillation maintained at resonant frequency, no instability
4	Phase isolation	CLK0 board has zero influence on CLK1 or CLK2 when all three powered
5	Labelling	Boards labelled X-AXIS / Y-AXIS / Z-AXIS matching CLK0 / CLK1 / CLK2